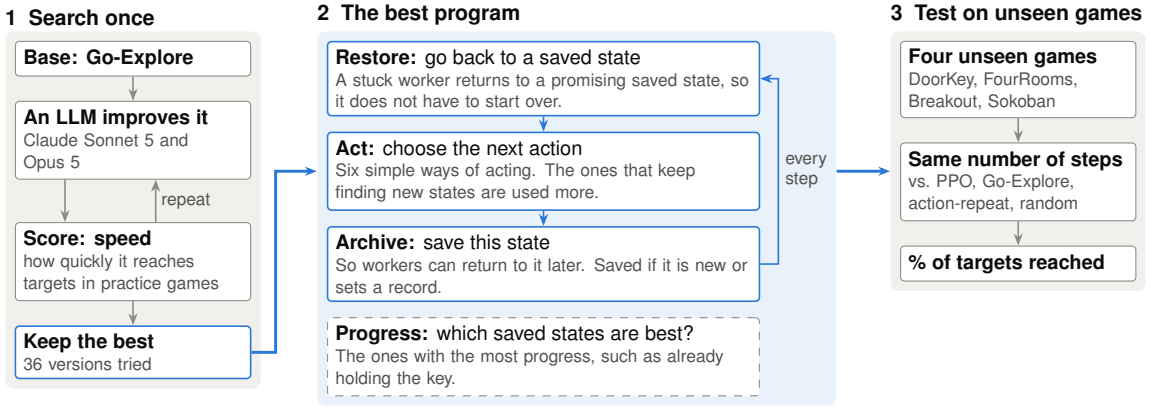
1 Search once
Base: Go-Explore
An LLM improves it
Claude Sonnet 5 and Opus 5
repeat
Score: speed
how quickly it reaches targets in practice games
Keep the best
36 versions tried
2 The best program
Restore: go back to a saved state
A stuck worker returns to a promising saved state, so it does not have to start over.
Act: choose the next action
Six simple ways of acting. The ones that keep finding new states are used more.
Archive: save this state
So workers can return to it later. Saved if it is new or sets a record.
every step
Progress: which saved states are best?
The ones with the most progress, such as already holding the key.
3 Test on unseen games
Four unseen games
DoorKey, FourRooms, Breakout, Sokoban
Same number of steps
vs. PPO, Go-Explore, action-repeat, random
% of targets reached